FIN 43900 / PROJECT 1

Research Evolution

IBM common equity
From initial thesis to investment committee decision

WATCH / DEFER

Edition A — September 1, 2026, 5:15–5:43 PM

Edition B — October 14, 2026

Valuation cutoff — September 1, 2026, after close

Audience — Fund Investment Committee

Decision unchanged. Research, modeling, valuation evidence, and risk qualification materially expanded.

Edition A • Written September 1, 2026, 5:15–5:43 PM. The content below is preserved from the original Edition A text provided for this project.

Project 1 — Edition A: IBM

Prepared for: Fund Investment Committee
Decision: Whether to allocate further diligence time toward a potential IBM equity investment. This is not yet a buy recommendation or authorization to trade.
Company / valuation date: International Business Machines Corporation (NYSE: IBM); valuation date **September 1, 2026**. The current share price, market capitalization, enterprise value, and trading multiples have not yet been collected, so no valuation conclusion is made in this edition.

Initial thesis — Watch / defer

IBM appears to have a potentially investable software-led earnings and cash-flow story: FY2025 revenue grew, Software revenue grew faster, and Software ARR increased. But the evidence collected so far does not establish whether those improvements are already reflected in IBM's share price, nor does it establish a satisfactory return after accounting for a meaningful debt balance. The appropriate initial posture is **watch / defer** pending valuation, cash-flow-quality, and leverage work.

Known evidence

All financial facts below are drawn from IBM's FY2025 Form 10-K, filed February 24, 2026 (accession **0000051143-26-000010**), including its incorporated 2025 Annual Report exhibit.

Evidence	FY2025 result	Why it matters
Total revenue	**$67.535bn** vs. $62.753bn in FY2024	Establishes renewed top-line growth of approximately 7.6%.
Software revenue	**$29.962bn** vs. $27.085bn	Software grew 10.6% reported, faster than consolidated revenue, supporting the mix-shift hypothesis.
Software ARR	**$23.6bn** at year-end, up about $2bn year over year	Recurring revenue is a relevant measure of software-business health and durability. IBM defines ARR as current-quarter recurring revenue multiplied by four.
Cash and cash equivalents	**$13.587bn**	Liquidity offset against debt, subject to further work on cash availability.
Debt	**$61.260bn** = $6.424bn short-term + $54.836bn long-term	Leverage is material and must be assessed against cash flow, interest burden, maturities, and IBM Financing.
Diluted EPS / shares	**$11.17** / **948.675m** weighted-average diluted shares	Establishes the per-share earnings base; it is not a substitute for a current diluted-share or enterprise-value bridge.

Source links: SEC filing detail; consolidated income statement; consolidated balance sheet; 2025 Annual Report, Exhibit 13.

Consequential assumptions

1. The increase in Software ARR reflects durable recurring demand rather than predominantly acquisition-related growth or a temporary booking effect.
2. IBM can service and refinance its debt without impairing shareholder returns or curtailing necessary investment.
3. Higher-growth software can increase IBM's earnings quality and valuation relevance even while Consulting and Infrastructure remain substantial parts of the business.
4. The FY2025 results are sufficiently representative to serve as a starting point for forward underwriting; this must be tested against FY2026 results and guidance.

Unknowns that prevent an initiation decision

- Current share price, market capitalization, enterprise value, and the multiples implied by the September 1, 2026 valuation date.
- Consensus revenue, EPS, free-cash-flow expectations, and the gap between those expectations and management guidance.
- A full cash-flow bridge: operating cash flow, capital expenditures, acquisitions, dividends, buybacks, and free cash flow after the chosen definition.
- Debt maturities, interest expense, floating/fixed-rate mix, restricted or overseas cash, and the relationship between operating debt and IBM Financing debt.
- Segment margins, organic versus acquired growth, retention/churn, customer concentration, competitive positioning, and the durability of ARR.
- The key catalyst path and the evidence that would show that the software-mix thesis is failing.

Research plan before returning to the committee

1. Capture IBM's closing share price and fully diluted share count as of September 1, 2026; build a market-cap-to-enterprise-value bridge.
2. Review FY2026 quarterly filings, earnings releases, and calls to update ARR, software growth, margins, free cash flow, guidance, and acquisition effects.
3. Build a three-case operating model separating Software, Consulting, Infrastructure, and Financing; distinguish organic growth from acquired revenue.
4. Build a debt and cash-flow schedule covering maturities, interest burden, lease obligations, IBM Financing, capital allocation, and deleveraging capacity.
5. Value IBM using forward EV/FCF and EV/EBIT approaches, with P/E only after validating EPS quality; compare the implied assumptions with relevant large-cap enterprise-software and IT-services peers.
6. Define the decisive KPI set and catalysts: Software ARR growth, software margin, free-cash-flow conversion, net debt trajectory, and forward guidance.

Partner-generated falsification question

If Software ARR growth slows materially while debt remains near FY2025 levels, what evidence would show that IBM's earnings multiple should compress rather than remain supported by its software mix?

Submission record

Checkpoint receipt identifier: Pending — the separate Project 1 Edition A checkpoint and lab checkout are not present in the local workspace. This file must remain unchanged once submitted.

Administrative note — October 14, 2026: The student verified a 25/25 Lab 03 attempt containing the complete original Edition A, but the separate ungraded Project 1 Edition A checkpoint currently shows no submission. This does not alter Edition A or establish checkpoint compliance. Status: **PENDING INSTRUCTOR DETERMINATION**; see `docs/edition-a-checkpoint-status.md`.

Edition B — Investment decision after research and model validation

Decision user: Fund Investment Committee
Security and valuation cutoff: International Business Machines Corporation (NYSE: IBM), September 1, 2026 after close
Research conclusion finalized: October 14, 2026
Horizon: 12–18 months of monitoring supported by a five-year operating forecast
Committee action: Watch / Defer; do not initiate a position at the cutoff price.

Current investment judgment

Edition A identified a potentially investable software-led business but could not determine whether its prospects were already priced in. Edition B finds that a stronger mix and margin story is possible, yet the model does **not** establish an adequate margin of safety at IBM's September 1 close of **$231.40**. The approved FCFF DCF values common equity at **$145.13 / $182.29 / $224.28 per diluted share** under low/base/high causal cases. The base case is **21.2% below** the reference price; even the high causal case is below it. The initial Watch / Defer recommendation remains, now for a valuation-based reason rather than a lack of a model.

Current valuation screen	Result
September 1 close / base per-share value	$231.40 / **$182.29**
Low / base / high causal DCF	$145.13 / $182.29 / $224.28
Base WACC / perpetual growth / incremental ROIC	7.07% / 2.5% / 15.0%
Operating enterprise value	$220.10bn
Terminal value as share of operating EV	**78.06%**
Reverse-DCF growth needed to match market price	**3.815%** at base WACC and 15% terminal ROIC
Quality-control record	**30/30 automated tests passing at Edition B finalization**; intentional $1m failure detected; published Week 3 benchmark reconciled. The later high-risk extension brings the current suite to 35 tests; see `docs/high-risk-validation-evidence.md`.

What the additional research established

Current operating evidence and the forecast. Research extended the FY2025 Form 10-K starting point to IBM's June 30, 2026 Form 10-Q and contemporaneous July earnings disclosure. FY2026 uses H1 reported actuals plus H2 forecast. Software growth is modeled at 6.0%, 5.5%, 5.0%, then 4.5% for 2027–2030; Consulting recovery and positive Infrastructure growth are less certain. Operating gross margin rises from 57.2% to 59.0% by 2030, while SG&A falls from 30.8% to 29.0% of operating revenue. These are *analyst forecasts*, not company guidance.

Integrated valuation, not just multiples. A connected five-year income statement, balance sheet, and cash-flow model supports an operating FCFF DCF. IBM Financing is excluded from operating cash flow and added once in the common-equity bridge using $1.450bn of reported June book equity. The model separately considers corporate debt, excess cash, minority interests and the declared-but-unpaid September 10 dividend. Oracle and Accenture EV/EBIT and P/E applications are **qualified reference anchors**, not interchangeable peers or a blended target.

Testing the assumptions. One-at-a-time operating shocks change intrinsic value by **+$12.20/share** for +1 percentage point of gross margin, **+$8.36** for +2 percentage points of Software growth, and **+$6.14** for a 0.5-point improvement in SG&A/revenue. These shocks have different magnitudes; their ordering is not a standardized ranking of economic importance. A favorable WACC / long-run-growth combination can put value above market, illustrating why the conclusion is conditional rather than certain.

Consequential repairs and disposition of AI assistance

The research process involved student-directed AI coding and analytical assistance, independent source and arithmetic checks, manual operating tests, and documented corrections. The important revisions were:

1. **Period boundary:** FY2026 balance-sheet cash and equity must roll forward from June 30 using H2 activity only, not a second application of H1. For the September 1 after-close DCF, remaining FY2026 FCFF begins September 2 and uses **121/184** of the H2 forecast, correcting a superseded **122/184** convention. Tests enforce the date and bridge logic.
2. **Equity bridge:** The $1.69 per-share dividend payable September 10 is deducted **once**; Financing operating and debt flows are not mixed into ex-Financing FCFF. A 952.7m H1 diluted-share average is explicitly a proxy, not a September 1 exact count.
3. **AI-output correction:** An earlier AI-assisted timing/valuation output used the incorrect stub convention. The discrepancy was checked against the valuation boundary, corrected in code, and captured in the AI-use record rather than accepted on authority.
4. **Stock-based compensation (SBC):** A subsequent review identified missing SBC treatment in forecast consolidated CFO. On October 14, the student approved **$938.5m H2 2026** and **$1.877bn each year 2027–2030** as provisional SBC assumptions. SBC was added once to CFO and APIC/equity, not added as free operating FCFF. The approved cash-bridge repair moved the **current base value from $181.95 to $182.29** without changing operating EV. Future dilution is not independently modeled.
5. **Historical locked experiment:** On October 10, the original *pre-SBC* workbench recorded a software-growth shift of 0 to +2 percentage points, with a pre-run prediction of higher revenue, EBIT, FCFF and value but no action change. Observed per-share value moved **$181.95 to $190.31** while Watch / Defer remained. Original BEFORE/AFTER text, commit references and screenshots are retained as historical evidence; that result is not restated as a test of the later $182.29 baseline.

What remains uncertain and what would change the decision

Material, unresolved cash-conversion qualification. After the SBC correction, the model's approximately **$10.918bn FY2026 company-style FCF proxy** is **$4.782bn lower** than a roughly $15.7bn management-outlook reference available by the cutoff. Forecast cash taxes, Financing-receivable movements, H2 seasonality, and the non-identical capital-investment definition remain insufficiently reconciled. Consolidated CFO, IBM-defined FCF and unlevered operating FCFF are **different measures**; the model was not forced to meet guidance.

Valuation qualification. The 0.70 beta has an opaque third-party methodology; the 5.47% pretax debt cost is a Treasury-plus-single-A-spread proxy rather than an IBM-specific bond yield; pension and Financing values require fuller reconciliation; future SBC dilution and point-in-time diluted shares remain unresolved. With **78.06%** of operating EV arising from terminal value, the DCF is sensitive to steady-state assumptions. The published Week 3 training-case reconciliation now passes all 12 published outputs within tolerance: conventional DCF value is **$27.4974/share at 10% WACC** and **$23.4140 at 11%**. It validates conventional DCF arithmetic only, not IBM-specific timing, Financing treatment, or ROIC-based terminal reinvestment; see `docs/week3-known-answer-reconciliation.md`.

Committee conditions. Maintain Watch / Defer until reported software demand, renewal performance, Consulting recovery and gross-margin/SG&A conversion support the forecast, and cash-flow quality and equity-bridge items become independently reproducible. Revisit after two relevant reporting quarters or sooner if credible evidence materially changes value. Re-underwrite downward if growth, conversion or leverage deteriorate, or if validated pension/dilution/financing claims reduce equity value. Initiation requires a **conservatively supported value above the prevailing price**, not merely an optimistic sensitivity cell.

Research responsibility and audit trail

The initial Edition A thesis and evidence were the student's baseline. The final forecasting choices, Watch / Defer action and approved SBC schedule are the student's investment judgments. AI-assisted code inspection, calculation repair, testing and document preparation are disclosed; no undocumented partner criticism is attributed to a third party. Primary-date sources and detailed tests are preserved in the repository's `evidence-ledger.md`, `docs/validation-and-ai-use.md`, `docs/stage2-investment-review.md`, `docs/stage2-sbc-review-approved-2026-10-14.md`, and `docs/locked-changed-input-record-*.txt`.

Edition A to Edition B — Evidence and decision evolution

Decision continuity: No material change in committee action—Watch / Defer in both editions. The evidence supporting that action changed materially: Edition A deferred because it lacked a valuation; Edition B deferred after valuing IBM below price, stress-testing the decision, and documenting unresolved accounting and source limitations.

Item	Edition A	Edition B	Primary driver of change	Evidence or test supporting the change (or no change)
Thesis and recommendation	Software-led earnings and ARR improvement; Watch / Defer pending valuation.	Watch / Defer after low/base/high $145.13/$182.29/$224.28 versus $231.40 market.	Own investment judgment + model-based analysis; AI-assisted implementation.	Current operating DCF, integrated cash-flow forecast, eight-slide deck; even high causal case is below price.
Information set and date discipline	FY2025 10-K, software ARR, debt and historical EPS; no cutoff price.	H1 2026 10-Q and July release, September 1 closing price, contemporaneous market inputs and peer records.	Own research; source-date validation.	`evidence-ledger.md`, FY2025 10-K and Q2 2026 10-Q, historical price records.
Financial model and architecture	Research plan for three cases, debt schedule, enterprise bridge and comps.	Connected FY2026 H1 actual + H2 forecast through FY2030 statements, FCFF DCF, IBM Financing sidecar, equity bridge and separate peer anchors.	Own research + AI-assisted coding and testing.	`model/`, `output/`, reconciled bridge, 30 automated tests, working browser app.
Timing and accounting repair	Full cash-flow bridge was an acknowledged unknown.	June cash/equity rolls using H2 only; post-close stub 121/184; September 10 dividend once; SBC add-back to CFO/APIC, not operating FCFF.	Mixed: independent date/source check, AI-assisted repair, student-approved SBC assumption.	Stub and dividend tests; FY2026 historical bridge; `stage2-sbc-review-approved-2026-10-14.md`.
Assumptions and sensitivity	Hypotheses about recurring software, refinancing and earnings quality without quantified sensitivities.	Segment growth, 57.2%-59% gross margin path, 7.07% WACC, terminal ROIC/reinvestment and named one-at-a-time tests.	Own forecast selection + computational tests.	Driver review, bounded live workbench, +$12.20 GM / +$8.36 Software OAT values; WACC/g grid.
Risk and uncertainty	Debt, cash-flow quality, organic growth and unknown market expectations.	78.06% terminal EV; qualified beta/debt yield, pension and dilution; $4.782bn management-style FCF gap.	Additional primary-source investigation and model challenge; AI-assisted diagnostic.	Reported versus proxy FCF bridge; independent source check; reverse DCF of 3.815% terminal g at market.
AI challenge and changed input	No completed model experiment or AI-based valuation to test.	Preserved pre-SBC locked software-growth +2pp experiment: $181.95 to $190.31, unchanged Watch / Defer. Later SBC repair separately approved and tested.	Student-authored prediction and evaluation; AI-assisted model/test development.	Original BEFORE/AFTER files, screenshot evidence and commit history; 30/30 tests at Edition B finalization, later extended to 35; `docs/week3-known-answer-reconciliation.md` and `docs/high-risk-validation-evidence.md`.
Decision conditions	Conduct more diligence before returning to committee.	Defer initiation; revisit after operating evidence and stronger cash conversion, WACC, pension and dilution support.	Own investment judgment, confirmed with model outcomes.	Base and named high cases below market, reverse DCF and explicit monitoring triggers.

Bottom line

The software-led IBM opportunity remains worth monitoring, but the research did not establish an attractive entry price on September 1, 2026. Edition B turns the initial research plan into a reproducible and falsifiable decision process. Watch / Defer is **confirmed by evidence**, not simply repeated from Edition A. The unresolved FCF guidance gap and financing/dilution/WACC limitations prevent presenting the $182.29 model result as a precise fair-value fact.

Primary audit artifacts: GitHub repository; `docs/Decision-Deck.pdf`; `docs/validation-and-ai-use.pdf`; `output/visible-base-output.html`; `docs/locked-changed-input-record-before.txt`; `docs/locked-changed-input-record-after.txt`. All market and filing data used for valuation are constrained to information available by the September 1, 2026 cutoff. Documentation and post-model repairs occurred afterward and are dated separately.